

Review of the corrected heavy-tail Section 5.2 refresh

SVP simulation audit

2026-09-15

Scope and acceptance checks

The regenerated study has 64000 method rows, 4 canonical algorithms, four patterns, 40 jump sizes, and 100 replications. The calibration uses 20000 calibration and 10000 validation null series. The nominal RFPOP target is 0.99 corrected null F1; the selected calibration estimate is 0.99615.

Table 1: Selected robust calibration parameters.

method	selected_parameter
RFPOP	3.000
Wilcoxon	1.750
MedianMood	0.001

The Gaussian-PELT external null-F1 benchmark is 0.987. The intentionally misspecified Gaussian PELT fit evaluated on the Student-t(2) validation stream has null F1 0.

Calibration table

The full candidate table, including Wilson intervals, seeds, target, and selection flags, is stored in `true_true_calibration.csv` in the robust study folder. The selected validation rows are shown here.

Table 2: Independent validation of selected null thresholds.

method	candidate	null_f1	null_f1_lower	null_f1_upper	rfpop_target_f1	selected	seed
RFPOP	3.000	0.9958	0.99433	0.99689	0.99615	TRUE	920000
Wilcoxon	1.750	0.9951	0.99353	0.99629	0.99615	TRUE	920000
MedianMood	0.001	0.9979	0.99679	0.99863	0.99615	TRUE	920000

Paired before/after comparison

The detailed comparison is in `paired_comparison.csv`, indexed by pattern, jump, and method. The table below reports the mean change over jump sizes for each pattern. Positive values improve the metric; for MSE and segment count, negative values are preferable.

Table 3: Mean new-minus-legacy metric change by pattern and method.

pattern	method	dF1	dPrecision	dRecall	dCorrect	dMSE	dSegments
none	PELT	-0.0258	-0.0131	-1.0000	0.0000	0.0000	0.0000
none	RFPOP	0.0476	0.0666	-0.0050	0.1105	-0.0068	-0.2020
none	SVP MedianMood	0.0041	0.0069	-0.0015	0.0153	-0.0006	-0.0155
none	SVP Wilcoxon	0.0031	0.0066	-0.0040	0.0173	-0.0014	-0.0173
rand1	PELT	-0.0202	-0.0117	-0.0421	0.0000	0.0000	0.0000
rand1	RFPOP	-0.0690	-0.0968	-0.0686	-0.0050	0.0008	-0.2902
rand1	SVP MedianMood	-0.0418	-0.1757	-0.0019	0.2297	-0.0895	0.8985
rand1	SVP Wilcoxon	-0.0537	-0.0510	-0.0565	-0.0365	-0.0100	-0.1928
up	PELT	-0.0202	-0.0117	-0.0417	0.0000	0.0000	0.0000
up	RFPOP	-0.1183	-0.1049	-0.1052	0.0000	0.0734	-0.2760
up	SVP MedianMood	-0.0397	-0.0910	-0.0215	0.1145	-0.1017	0.4695
up	SVP Wilcoxon	-0.0620	-0.0634	-0.0609	-0.0285	-0.0261	-0.0975
updown	PELT	-0.0202	-0.0117	-0.0413	0.0000	0.0000	0.0000
updown	RFPOP	-0.0858	-0.1698	-0.0746	-0.0400	0.0060	-0.6450
updown	SVP MedianMood	-0.0376	-0.2366	-0.0057	0.1350	-0.0609	0.8145
updown	SVP Wilcoxon	-0.0633	-0.0647	-0.0622	-0.0312	0.0011	-0.1077

Revised Section 5.2 wording

Section 5.2 should describe the Student-t(2) experiment as a stress test with the fixed Gaussian PELT baseline and calibrated robust procedures. All noise is generated by `ts_generator(type = "student", df = 2, scale = 1)` for `n=1000`, four signal patterns, jump sizes `0.1, ..., 4`, and 100 replications. The endpoint `n` is excluded before one-to-one matching within the framework tolerance. Precision, recall, F1, correct-number probability, MSE, segment count, and localization error are then computed from the same metric code as the Gaussian study.

RFPOP uses biweight loss (the `robseg` name is `loss = "Outlier"`) with `lthreshold = 3` and `lambda = c_RF log(n)`. Its multiplier is calibrated on independent t(2) null streams for at least 0.99 corrected null F1. The Median-Mood and Wilcoxon SVP thresholds are independently selected to match that RFPOP target and use `subtests = "right"` only. Their K-dependent formulas use `oracle_segments`, the known data-generating number of segments in this power experiment; this oracle is not available in ordinary applications.